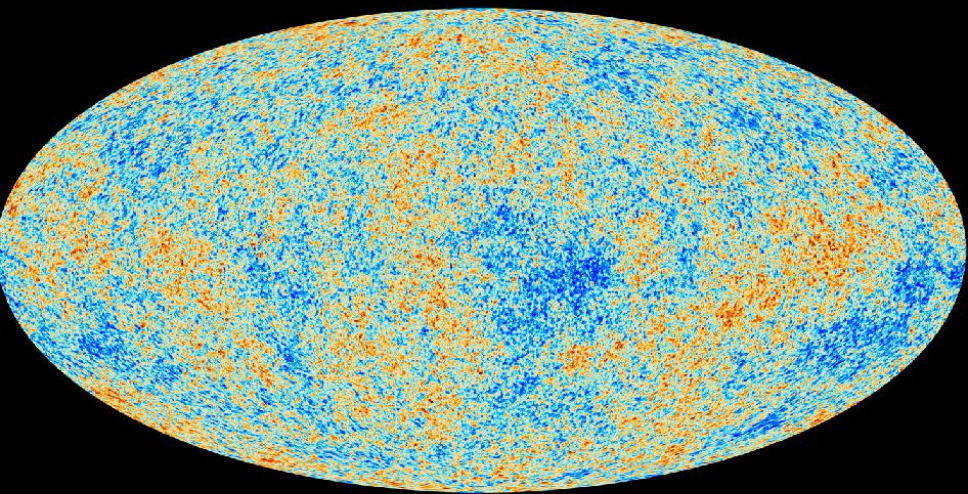




The Cosmic Microwave Background

Our probe of the birth of the universe

Will Handley

wh260@cam.ac.uk

Astrophysics Department
Cavendish Laboratory
University of Cambridge

2nd May 2017

Overview

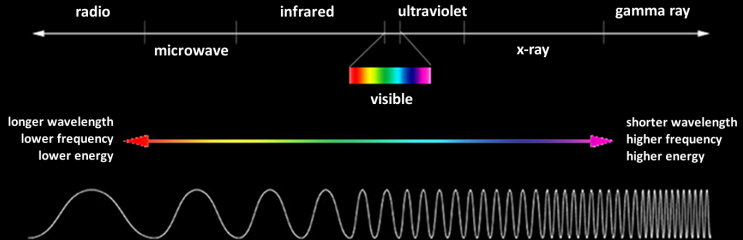
Light in astronomy

The universe in microwaves

How do we measure the CMB

What does the CMB tell us

The Electromagnetic Spectrum



Redshift

- ▶ Distant objects are *redder* than they should be.

Redshift

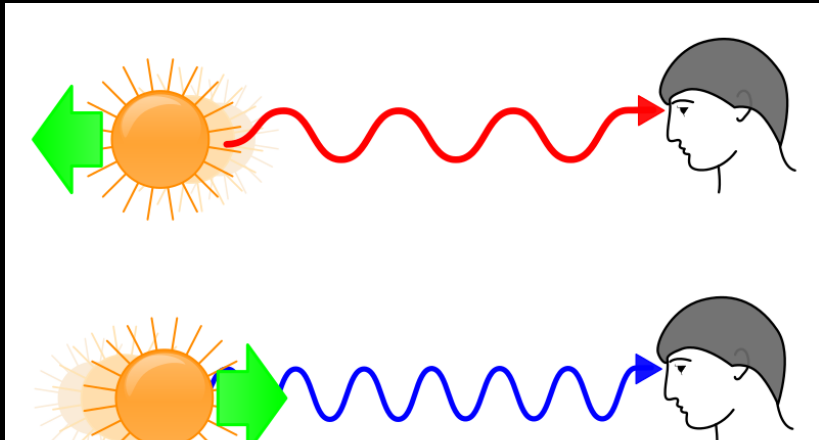
- ▶ Distant objects are *redder* than they should be.
- ▶ Two things can cause this:

Redshift

- ▶ Distant objects are *redder* than they should be.
- ▶ Two things can cause this:
 1. Doppler shift from objects moving away

Redshift

- ▶ Distant objects are *redder* than they should be.
- ▶ Two things can cause this:
 1. Doppler shift from objects moving away

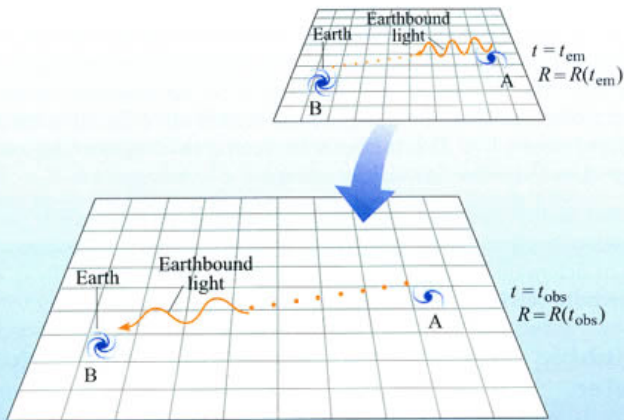


Redshift

- ▶ Distant objects are *redder* than they should be.
- ▶ Two things can cause this:
 1. Doppler shift from objects moving away
 2. Stretching by the expansion of the universe.

Redshift

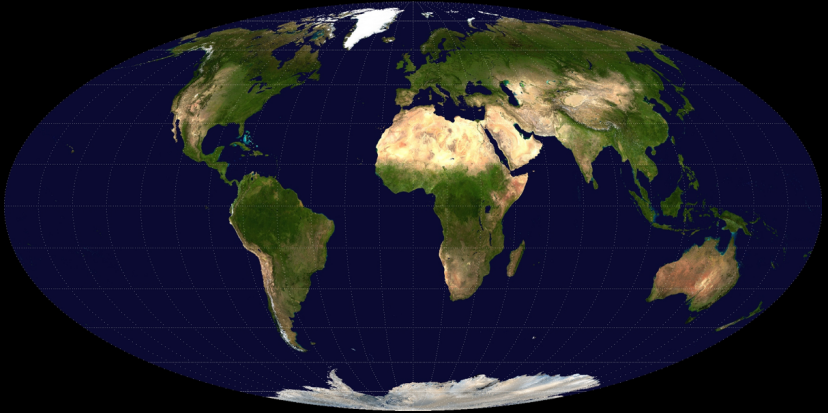
- ▶ Distant objects are *redder* than they should be.
- ▶ Two things can cause this:
 1. Doppler shift from objects moving away
 2. Stretching by the expansion of the universe.



Redshift

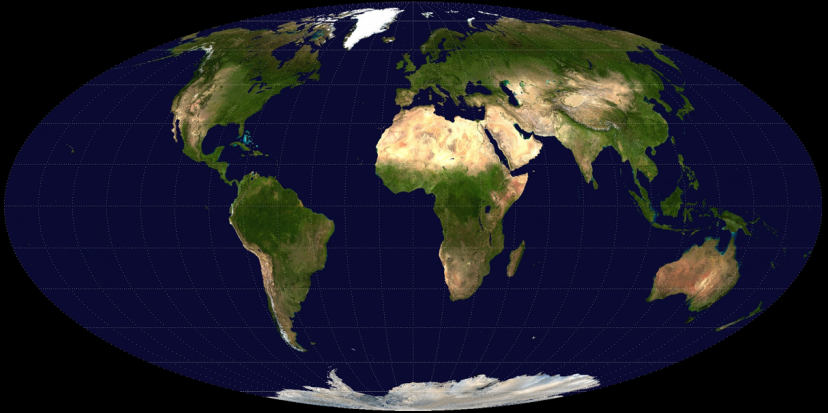
- ▶ Distant objects are *redder* than they should be.
- ▶ Two things can cause this:
 1. Doppler shift from objects moving away
 2. Stretching by the expansion of the universe.
- ▶ Note; the redshift of distant galaxies is often (incorrectly) attributed to 1.

The Cosmic Microwave Background



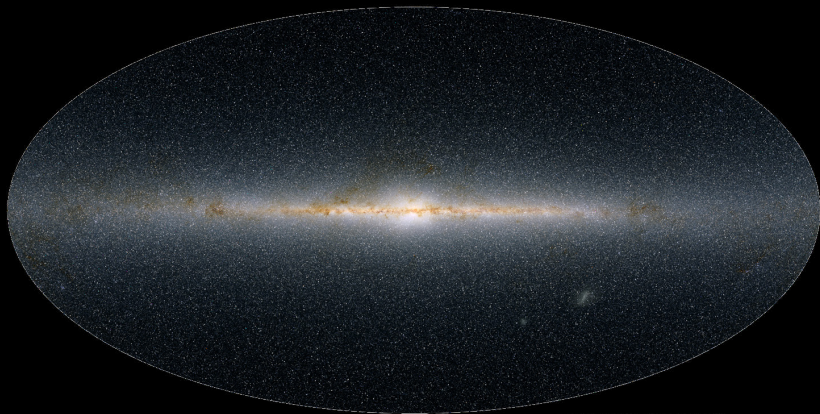
- ▶ A map of the Earth (Mollweide projection).

The Cosmic Microwave Background



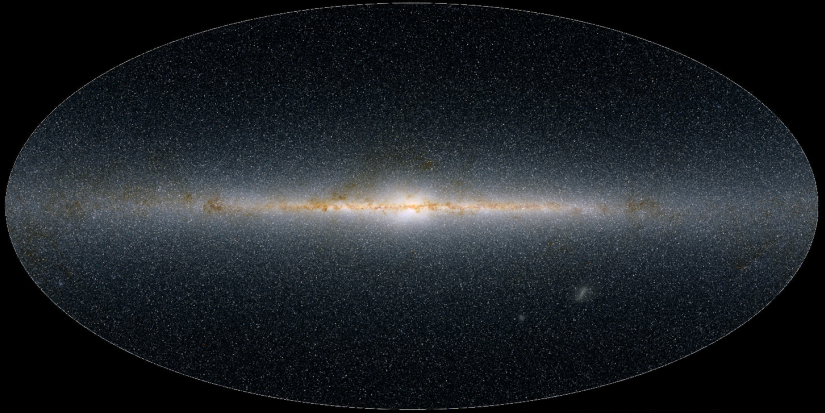
- ▶ A map of the Earth (Mollweide projection).
- ▶ Consider the same map, but looking out at the night sky.

The Cosmic Microwave Background



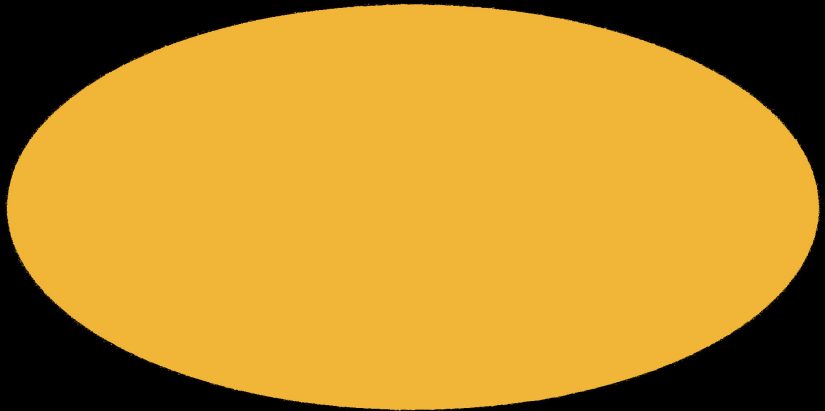
- This image is taken in the infra-red waveband.

The Cosmic Microwave Background



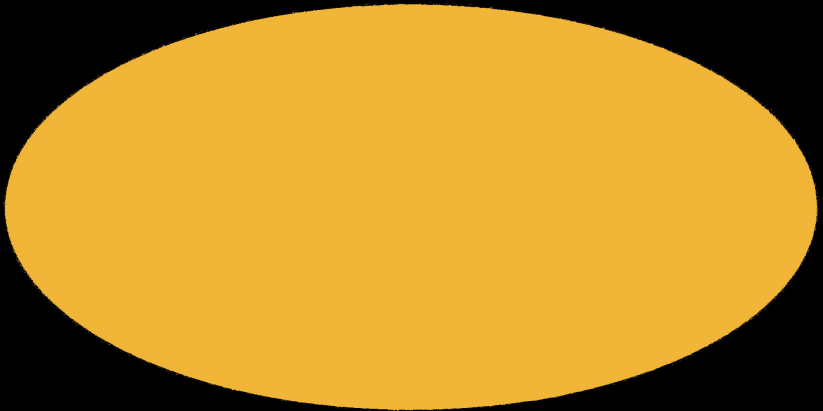
- ▶ This image is taken in the infra-red waveband.
- ▶ Now look in the microwave band.

The Cosmic Microwave Background



- ▶ Now look in the microwave band.

The Cosmic Microwave Background



- ▶ Now look in the microwave band.
- ▶ The sky is filled with the Cosmic Microwave Background radiation (CMB), coming in from all directions.

The Cosmic Microwave Background

Discovery

- ▶ Discovered by accident in 1964 by Penzias & Wilson.



The Cosmic Microwave Background

Discovery

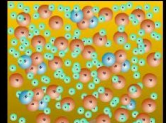
- ▶ Discovered by accident in 1964 by Penzias & Wilson.
- ▶ Experimenting with a supersensitive detector.



The Cosmic Microwave Background

What is it?

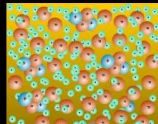
- ▶ ~ 13.8 billion years ago the universe was much smaller.



The Cosmic Microwave Background

What is it?

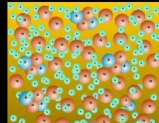
- ▶ ~ 13.8 billion years ago the universe was much smaller.
- ▶ The entire universe was as dense and hot as a star.



The Cosmic Microwave Background

What is it?

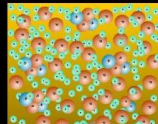
- ▶ ~ 13.8 billion years ago the universe was much smaller.
- ▶ The entire universe was as dense and hot as a star.
- ▶ Plasma: Filled with unbound protons, neutrons, electrons and photons.



The Cosmic Microwave Background

What is it?

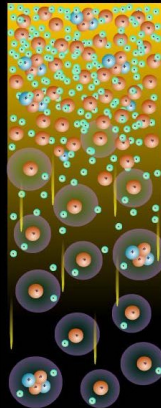
- ▶ ~ 13.8 billion years ago the universe was much smaller.
- ▶ The entire universe was as dense and hot as a star.
- ▶ Plasma: Filled with unbound protons, neutrons, electrons and photons.
- ▶ As the universe expanded, it cooled and eventually released these photons at the *epoch of recombination*.



The Cosmic Microwave Background

What is it?

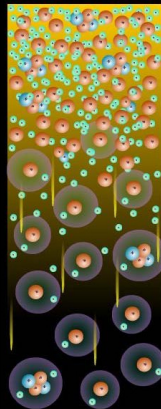
- ▶ ~ 13.8 billion years ago the universe was much smaller.
- ▶ The entire universe was as dense and hot as a star.
- ▶ Plasma: Filled with unbound protons, neutrons, electrons and photons.
- ▶ As the universe expanded, it cooled and eventually released these photons at the *epoch of recombination*.



The Cosmic Microwave Background

What is it?

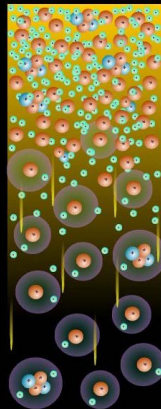
- ▶ ~ 13.8 billion years ago the universe was much smaller.
- ▶ The entire universe was as dense and hot as a star.
- ▶ Plasma: Filled with unbound protons, neutrons, electrons and photons.
- ▶ As the universe expanded, it cooled and eventually released these photons at the *epoch of recombination*.
- ▶ “Let there be light” (X rays).



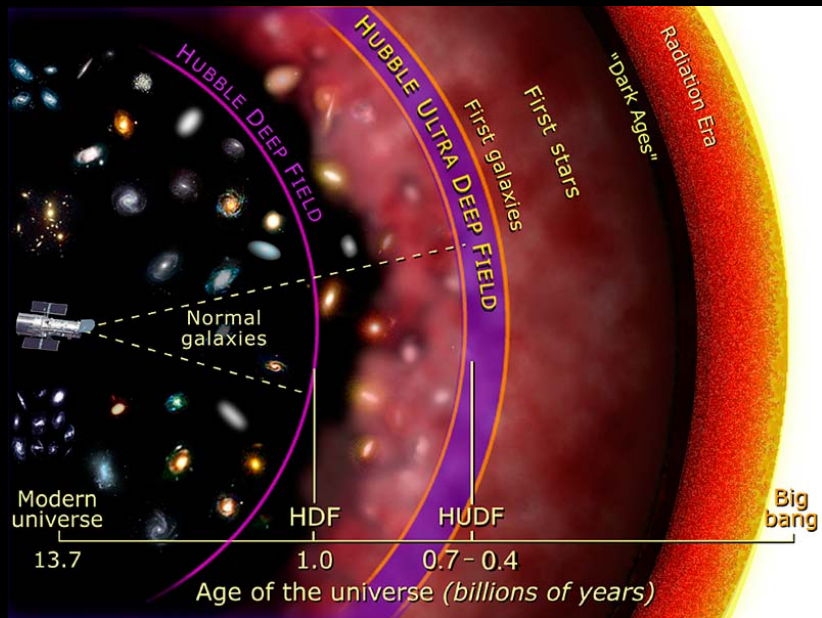
The Cosmic Microwave Background

What is it?

- ▶ ~ 13.8 billion years ago the universe was much smaller.
- ▶ The entire universe was as dense and hot as a star.
- ▶ Plasma: Filled with unbound protons, neutrons, electrons and photons.
- ▶ As the universe expanded, it cooled and eventually released these photons at the *epoch of recombination*.
- ▶ “Let there be light” (X rays).
- ▶ These X-rays have now been stretched to microwaves.

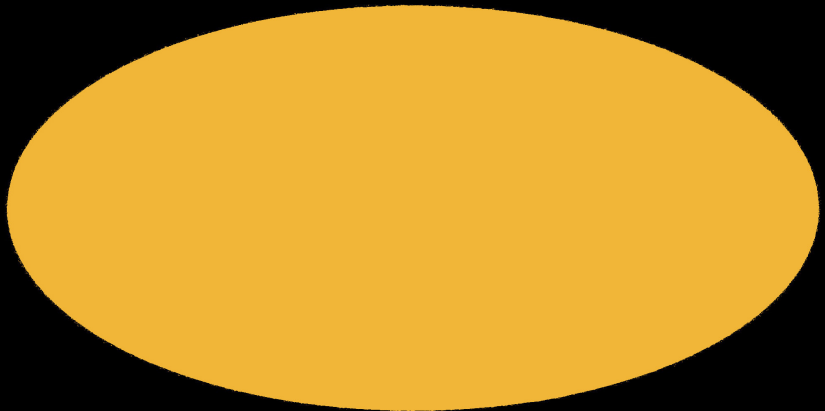


An Alternative Picture



Anisotropies in the CMB

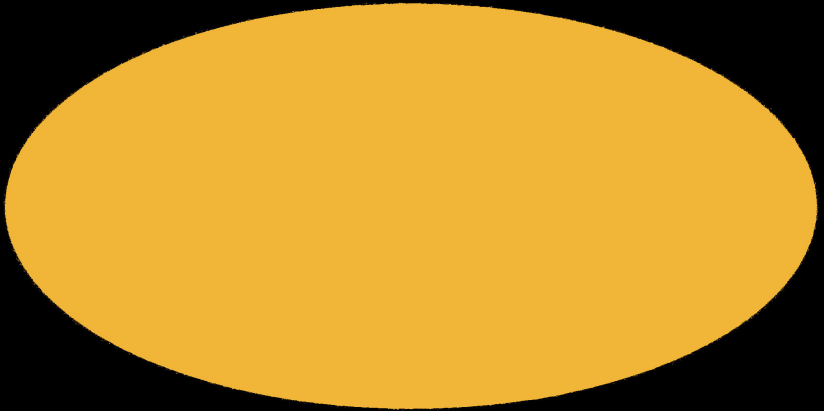
Looking for a more interesting picture



- ▶ Try to 'subtract off' the constant source.

Anisotropies in the CMB

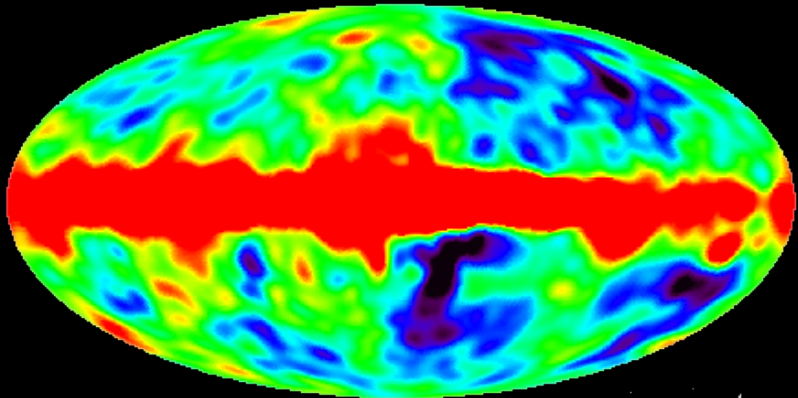
Looking for a more interesting picture



- ▶ Try to 'subtract off' the constant source.
- ▶ COBE microwave satellite (1992)

Anisotropies in the CMB

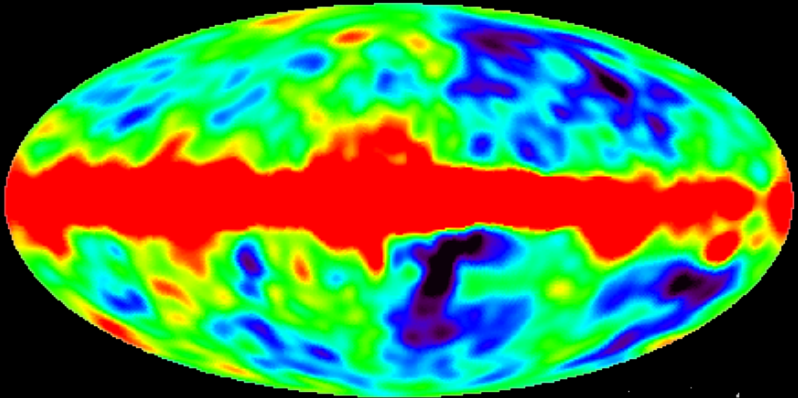
Looking for a more interesting picture



► COBE's view of the universe

Anisotropies in the CMB

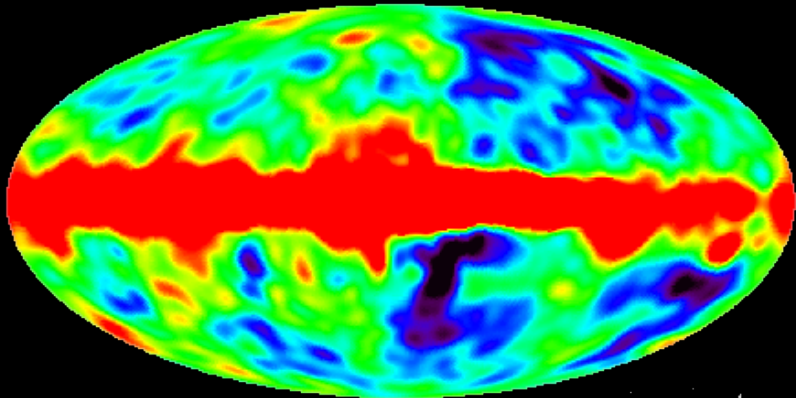
Looking for a more interesting picture



- ▶ COBE's view of the universe
- ▶ These direction-dependent deviations are called *Anisotropies*

Anisotropies in the CMB

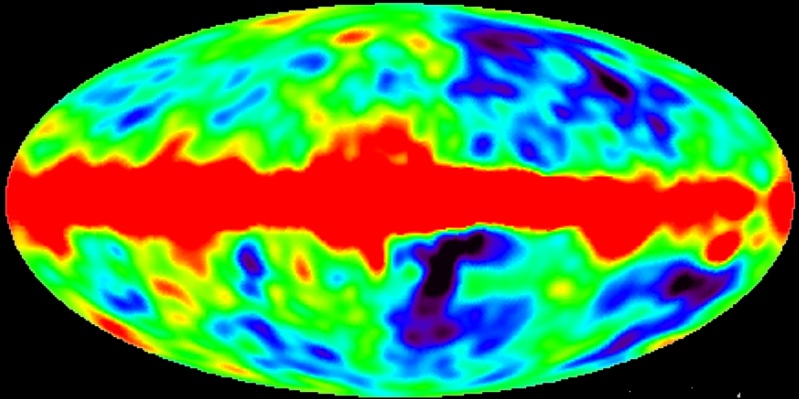
Looking for a more interesting picture



1. Galaxy in the middle

Anisotropies in the CMB

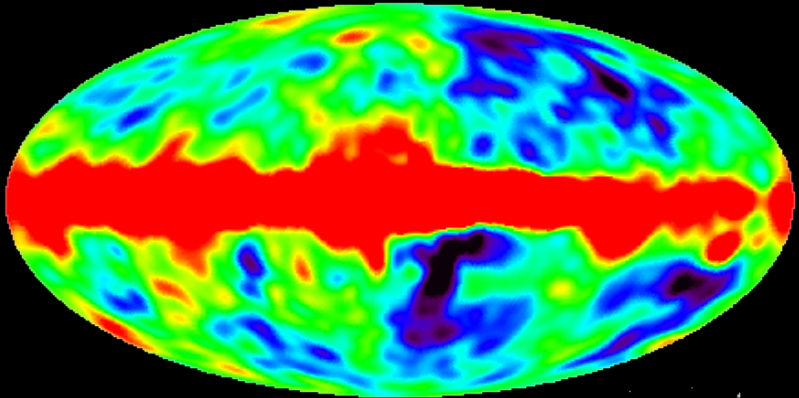
Looking for a more interesting picture



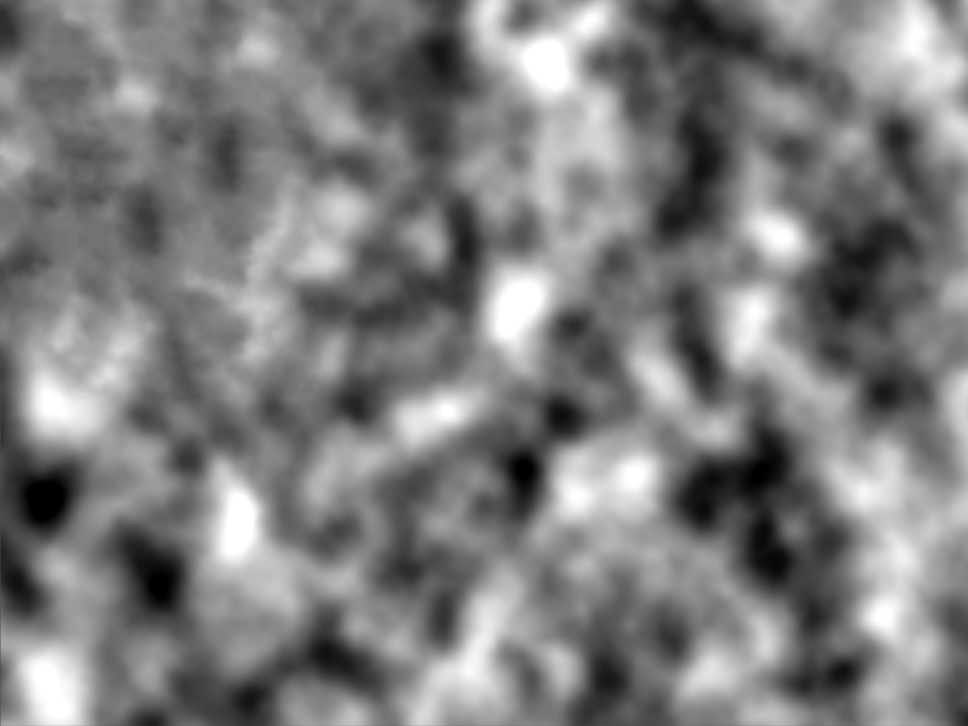
1. Galaxy in the middle
2. One half hotter than the other

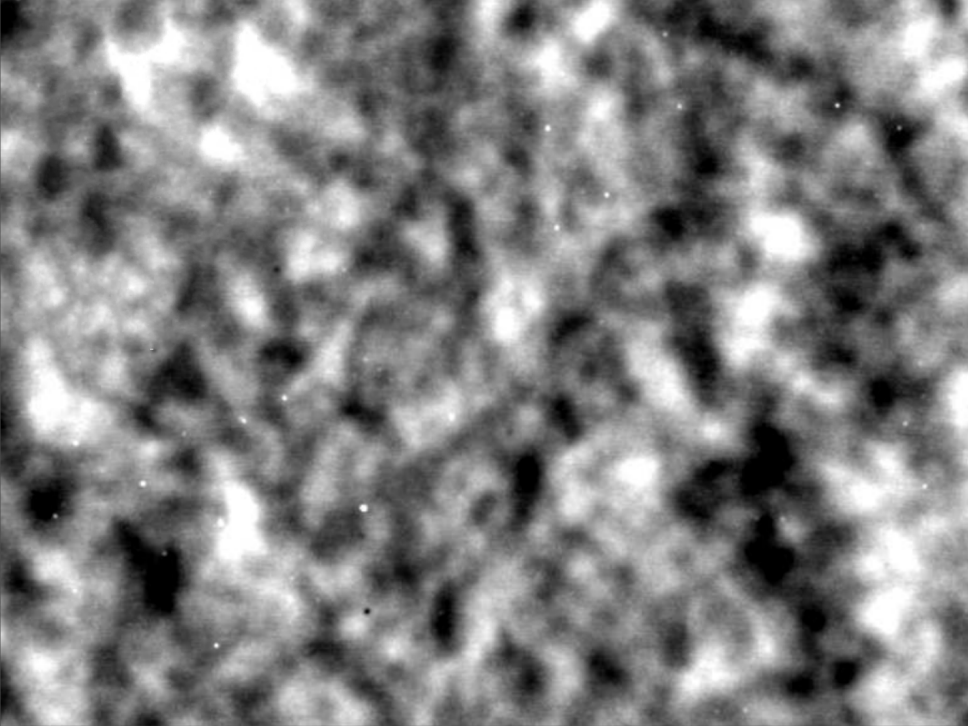
Anisotropies in the CMB

Looking for a more interesting picture



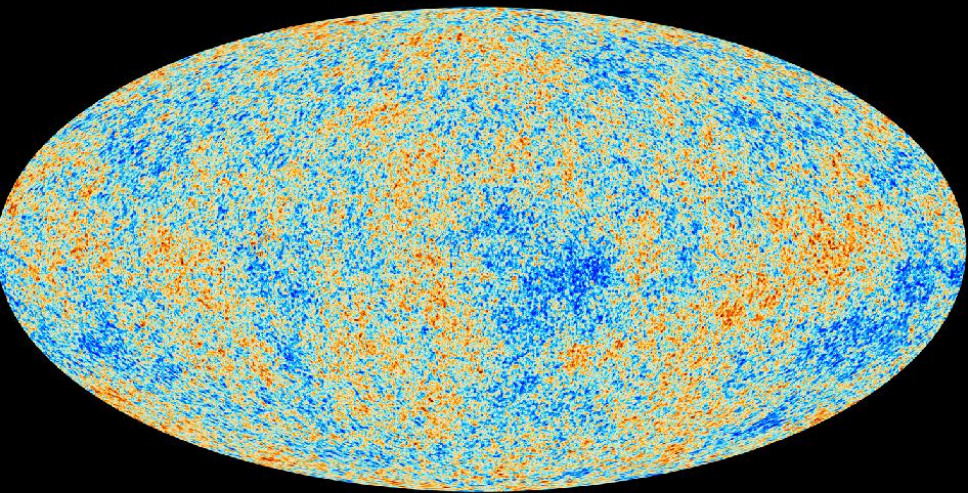
1. Galaxy in the middle
2. One half hotter than the other
3. Additional fluctuations





The Beginning of time

As seen by the Planck Satellite



How do we measure the CMB?

Three methods

1. From the ground.

How do we measure the CMB?

Three methods

1. From the ground.
2. From space.

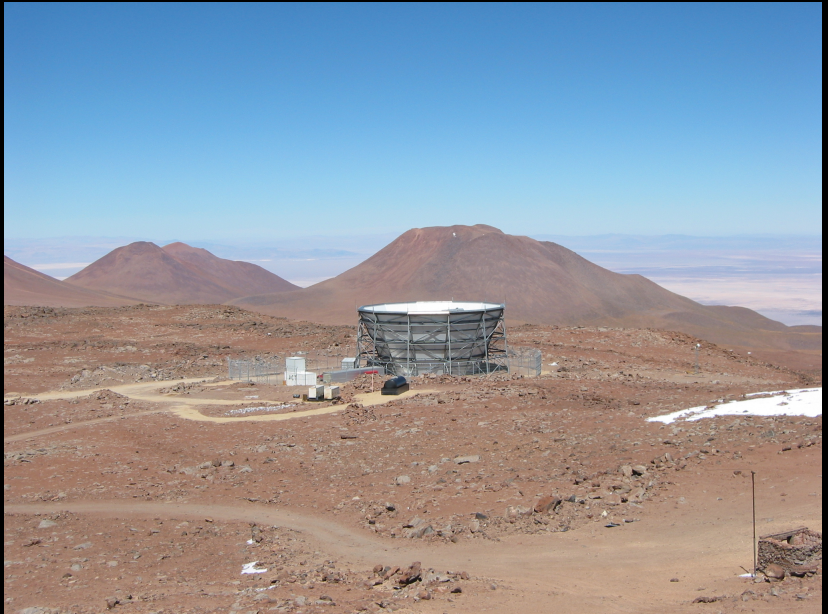
How do we measure the CMB?

Three methods

1. From the ground.
2. From space.
3. From balloons.

From the ground

Atacama Cosmology Telescope (ACT)



From the ground

South Pole Telescope (SPT)



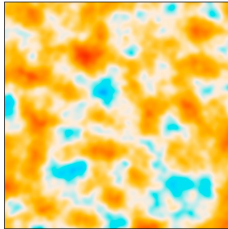
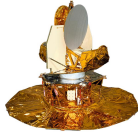
South Pole Telescope

From Space

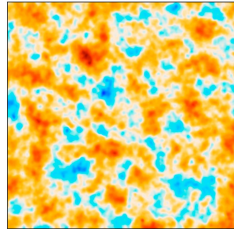
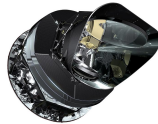
COBE, WMAP, Planck



COBE



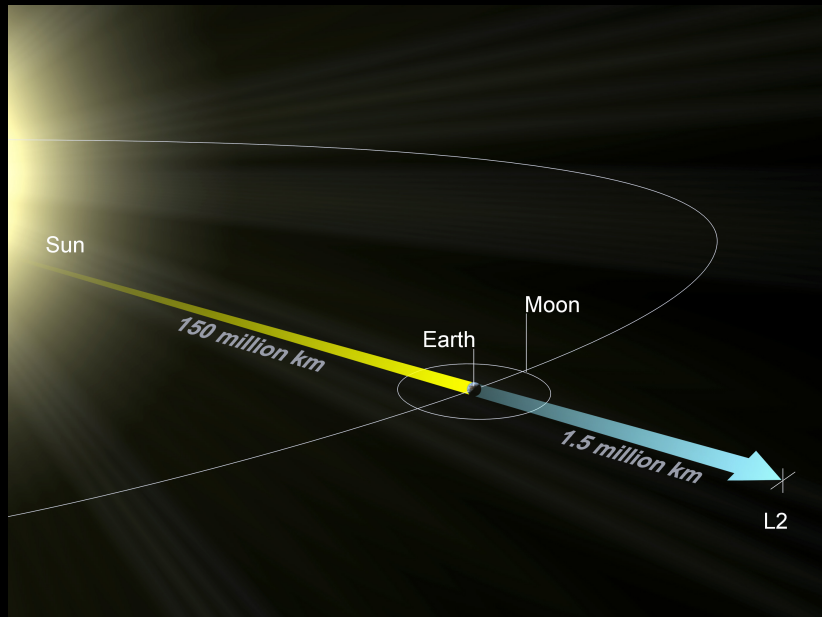
WMAP



Planck

From Space

COBE, WMAP, Planck



From Balloons



What does the CMB tell us?

- ▶ The CMB is a crystal clear picture of the universe when it was only 300,000 years old (epoch of recombination).

What does the CMB tell us?

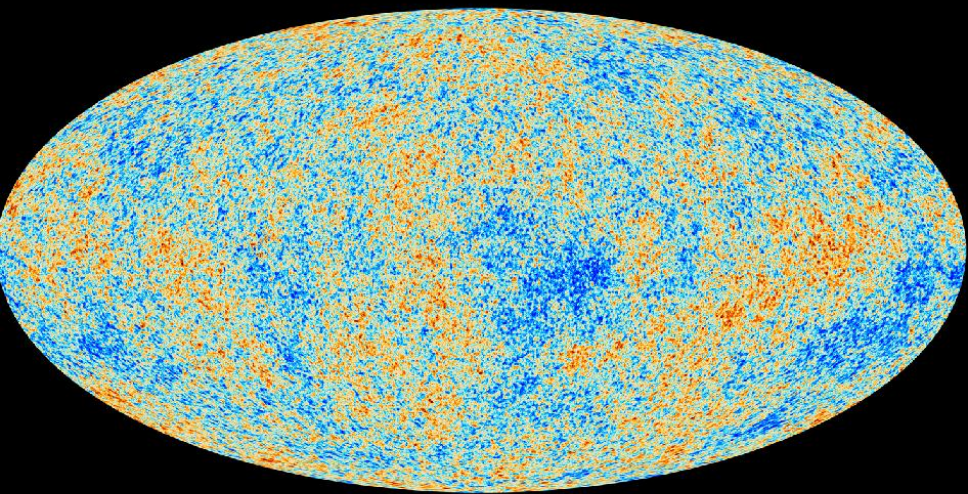
- ▶ The CMB is a crystal clear picture of the universe when it was only 300,000 years old (epoch of recombination).
- ▶ It therefore provides a unique insight into the birth of the universe.

What does the CMB tell us?

- ▶ The CMB is a crystal clear picture of the universe when it was only 300,000 years old (epoch of recombination).
- ▶ It therefore provides a unique insight into the birth of the universe.
- ▶ It gives crucial data to suggest and rule out theories of the beginning of time.

What does the CMB tell us?

- ▶ The CMB is a crystal clear picture of the universe when it was only 300,000 years old (epoch of recombination).
- ▶ It therefore provides a unique insight into the birth of the universe.
- ▶ It gives crucial data to suggest and rule out theories of the beginning of time.
- ▶ More than this, modern astrophysics uses the CMB as a “back light” for the dark regions of the universe.



What does the CMB tell us?

- ▶ The CMB is a crystal clear picture of the universe when it was only 300,000 years old (epoch of recombination).
- ▶ It therefore provides a unique insight into the birth of the universe.
- ▶ It gives crucial data to suggest and rule out theories of the beginning of time.
- ▶ More than this, modern astrophysics uses the CMB as a “back light” for the dark regions of the universe.